

Ramesh Aravindh T

Full-Stack Developer

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SUMMARY

Focused on building production applications, data-driven solutions, and intelligent systems, combining software development, data analytics, and applied AI to solve practical business problems.

EXPERIENCE

Full-Stack Developer

May 2025 — Present

Redlitmus Communications · India

- Built Jumbo Quiz, a Django 5 + DRF platform that turns uploaded PDF, DOCX, and spreadsheet coursework into ready question papers via a LangGraph + OpenAI pipeline with pgvector retrieval, replacing manual paper-setting.
- Built MeterSquare ERP on FastAPI and Supabase Postgres — costing, procurement, material and labour tracking, payroll, and maintenance under one nine-role approval hierarchy with value-threshold gates and row-level security.
- Delivered Office Task Manager, a real-time telecom task and ticketing platform with change-data-capture sync, optimistic UI with rollback, SLA escalation that auto-classifies and freezes overdue tasks, and an AI assistant built on pgvector RAG.

Python Developer Intern

Sep 2024 — Apr 2025

Nobel Software · India

- Built practical Python and SQL foundations through hands-on training, and helped newer interns work through core concepts.

SELECTED PROJECTS

Office Task Manager — Real-Time Telecom Ops Platform & AI Assistant

React 18 · TypeScript · Flask-SocketIO · gevent · PostgreSQL · pgvector · Supabase

- Ran Socket.IO in gevent mode with pycogreen-cooperative pycpg2 so one slow query cannot freeze the event loop, keeping websockets alive across full LLM round-trips.
- Bounded the AI assistant's natural-language-to-SQL path with a sqlparse table allow-list and user-scoped CTE rewriting — a parser-level boundary, not prompt instructions.

MeterSquare ERP — Construction ERP & Approval Hierarchy

FastAPI · Python · Supabase · PostgreSQL · SQLAlchemy

- Unified costing, procurement, material and labour tracking, payroll, and maintenance under one role hierarchy with value-threshold approval gates, enforced by Postgres row-level security rather than per-route checks.

Valoryx Software — Offline-First GST Billing & Business Management Platform

valoryx.ryxtech.in

React · TypeScript · Flask · SQLAlchemy · Supabase · SQLite · Electron

- Built an offline-first billing platform covering GST invoicing, inventory, multi-branch operations, business reporting, and thermal printing, with multi-tenant isolation on every query and TOTP 2FA.
- Cut page loads from 3–9 seconds to 1–20 ms — a 100×+ improvement on hot paths — by replacing per-request Supabase round-trips with a local SQLite mirror that keeps the counter running through network outages and syncs on reconnect.
- Shipped one codebase as both a responsive web app and an auto-updating Electron desktop build, with dashboards for sales and operational performance.

- Architected 26 domain-bounded Django apps over 129 multi-tenant models, exposing 280+ OpenAPI-documented REST routes across nine role-scoped dashboards — covering admissions, academics, attendance, fees, examinations, transport, library, and HR/payroll.
- Implemented tenant isolation, RBAC, JWT and OTP authentication, TOTP 2FA, and PostgreSQL row-level security for school-level data separation.
- Built asynchronous AI workflows on Celery and Redis for question generation, exam-blueprint creation, online assessment, and written-answer grading.

Uber Supply-Demand Gap Analysis — Python EDA & BI

github.com/rameshsurya10/Uber-Supply-Demand-Gap-EDA

Python · Pandas · NumPy · Matplotlib · Seaborn · Power BI

- Analysed ride-request, cancellation, and fulfilment patterns to isolate peak-demand periods and the locations where supply shortages concentrate.
- Cleaned and explored trip-level data with time- and location-based segmentation, then reported through Power BI with recommendations for driver allocation and availability.

Alumni Career Analysis — SQL Segmentation & BI Dashboards

github.com/rameshsurya10/Alumni-Career-Analysis

SQL · MySQL · PostgreSQL · Python · Pandas · Power BI · Tableau

- Used SQL and exploratory analysis to examine career paths, industry distribution, job roles, and salary progression across alumni records.
- Built analytical views and visualisations answering business questions on industry growth, career progression, and compensation trends.

TECHNICAL SKILLS

Backend: Python · Django · Django REST Framework · Flask · FastAPI · SQLAlchemy · Celery · Socket.IO · REST APIs · JWT · OAuth2

Frontend: TypeScript · JavaScript · HTML · CSS · React · Next.js · Vite · Redux Toolkit · TanStack Query · Tailwind CSS · Material UI

AI & Data: OpenAI API · LangChain · LangGraph · RAG · pgvector · Pandas · NumPy · Power BI · Tableau

Databases & Infrastructure: SQL · PostgreSQL · MySQL · SQLite · Redis · Supabase · Neo4j · Docker · AWS S3 · Linux · Git

EDUCATION & CERTIFICATIONS

Post Graduate Programme, Data Analytics & Data Science — Skill Lync, India 2023 — 2024

B.E. Computer Science and Engineering — RVS College of Engineering and Technology, Coimbatore, India 2019 — 2023

Certifications: Python Programming · MySQL · Power BI · Tableau · Advanced Excel